

VERSION-2

INVENTORY MANAGEMENT SYSTEM (IMS)

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Inventory Management System (IMS)

Needs Assessment & Business Case

1. **Introduction / Background:** The Inventory Management System (IMS) project is planned to fix problems in how businesses manage stock, track products, and keep inventory records. Many companies still use manual methods or separate systems, which can cause mistakes, delays, and higher costs. IMS will give a single, reliable solution for managing inventory.

2. Business Goals

- Make inventory records more accurate
- Reduce stock shortages and extra stock
- Improve work efficiency
- Help make better decisions with up-to-date inventory information

3. **Current Situation and problem/opportunity statement:** Tracking inventory is slow and often has mistakes because records are kept by hand and not updated in real time. This is a chance to create an automated system that saves time, reduces losses, and makes customers happier.

4. Critical Assumptions and Constraints

- Users know basic computer skills
- Necessary computers and internet are available
- Management supports the project

Limits: - Small project budget - Fixed timeline (school/academic schedule) - Small development team

5. Analysis Options and Recommendation

- Option 1: Keep using manual inventory (cheap but risky)
- Option 2: Buy ready-made software (medium cost, less flexible)
- Option 3: Build a custom IMS (more work, best fit)

Recommendation: Option 3 is recommended because it meets the business needs and can be customized.

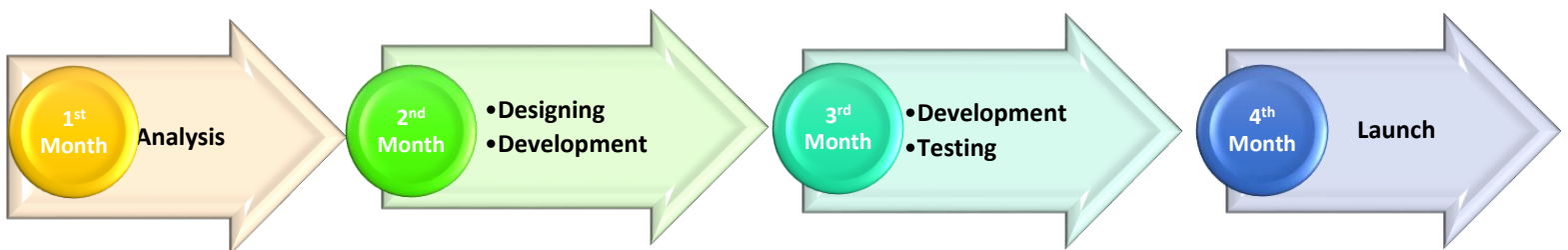
6. Preliminary Project Requirements

- Manage products and stock
- Record stock coming in and going out

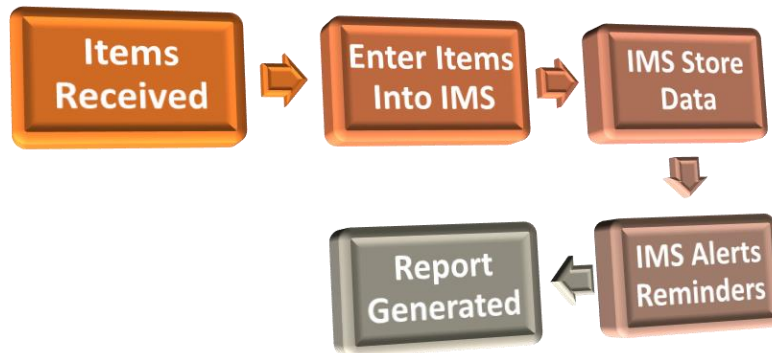
- Send alerts for low stock
 - Provide simple reports
7. **Budget Estimate:** The estimated cost covers software development, testing, and launching the system. The cost is moderate, good for a small-to-medium project.
8. **Timeline Estimate:** The project should take about 3–4 months, including analysis, design, development, testing, and launch.
9. **Possible Risks**
- Users may resist changes
 - Delays in the schedule
 - Requirements might not be clear

10. Exhibits

Project Timeline (3–4 Months)



System Over View



Identifying and Engaging Stakeholders

11. Stakeholder Register

Stakeholder Name	Contact Information	Project Role	Project Requirements	Project Concerns	Influence
John Mugisha	+900 588 123 456 jmugisha@gmail.com	Project Sponsor	System delivered on time and within budget	Cost overruns, delays	Supportive
Alice Uwimana	+901 722 456 7 auwimana@gmail.com	Project Manager	Clear scope, smooth execution	Schedule delays	Supportive
Patrick Ndayishimiye	+902 733 987 6 pndayi@gmail.com	Inventory Manager	Accurate stock tracking, expiry alerts	Incorrect stock data	Supportive
Jean Claude Habimana	+903 780 654 3 jhabimana@gmail.com	Store Staff	Simple and fast data entry	System too complex	Neutral
Beata Mukamana	+904 729 111 2	Procurement Officer	Low-stock alerts, reorder reports	Late purchase decisions	Supportive

	bmkana@gmail.com				
Eric Bizimana	0783 444 555 ebizi@gmail.com	Finance Officer	Correct inventory valuation reports	Inaccurate costing	Neutral
Samuel Karegeya	5730 999 888 skare@gmail.com	IT Administrator	Secure system, backups	Data loss, system failure	Neutral
Dev Team Lead	devteam@ims.com	Development Team	Clear requirements	Changing requirements	Supportive
Jean Claude	claudelj@ims.com	Senior Manger	Project aligns with company goals	Cost, Risk	Supportive

12. Stakeholder Mapping

	High Interest	Low Interest
High Power	Project Sponsor, Project Manager	Senior Management
Low Power	Inventory Manager ,Store Staff , Procurement Officer, Development Team	Finance Officer, ITAdministrator

Forming The Team

Team members

Name	Role
John Mugisha	Project Sponsor
Alice Uwimana	Project Manager
Sarah Njeri	Business Analyst
David Kimani	Lead Developer
Brian	Software Developer
Grace	UI Designer
Michael	Quality Assurance

RACI Chart

Activity	Project Sponsor	Project Manager	Business Analyst	Lead Dev	Soft Dev	UI	QA
Develop Project Charter	I	A	C	I	I	I	I
Define Project Scope	I	A	R	C	I	I	I
Gather Requirements	I	A	R	C	I	I	I
Approve Requirements	I	R	C	I	I	I	I
System Architecture Design	I	I	C	A	C	I	I
UI/UX Design	I	I	C	C	I	A	I
System Development	I	I	C	A	R	I	I
8. System Testing		I	C	C	I	I	A
9. User Acceptance Testing	I	A	R	I	I	I	C
10. Deployment & Project Closure	I	A	C	C	I	I	C

R = Responsible **A** = Accountable **C** = Consult **I** = Informed

TEAM CHARTER

Team Purpose

The purpose of this team is to successfully design, develop, test, and deploy an Inventory Management System (IMS) that improves stock tracking accuracy, reduces shortages and excess inventory, and enhances operational efficiency.

Team Objectives

- Deliver a functional IMS within the approved timeline

- Meet all defined requirements
- Ensure system reliability and usability
- Maintain effective communication among team members
- Deliver quality documentation

Roles and Responsibilities

- **Project Manager** – Oversees planning, scheduling, risk management, stakeholder communication.
- **Business Analyst** – Gathers, documents, and validates requirements.
- **Lead Developer** – Designs system architecture and supervises development.
- **Software Developer** – Implements system features.
- **UI/UX Designer** – Designs system interface and user experience.
- **QA/Test Engineer** – Conducts testing and ensures system quality.

Communication Plan

- Weekly team meeting (every Monday)
- Progress updates via email or WhatsApp group
- Urgent issues communicated immediately
- Formal reports submitted to Project Sponsor bi-weekly

Decision-Making Approach

- Technical decisions → Led by Lead Developer
- Requirement decisions → Led by Business Analyst
- Strategic decisions → Approved by Project Manager
- Major changes → Approved by Project Sponsor

Conflict Resolution

- Issues discussed internally first
- If unresolved → Escalated to Project Manager
- If still unresolved → Sponsor intervention

GROUND RULES

1. All members must attend scheduled meetings.
2. Deadlines must be respected.
3. Members must communicate blockers within 24 hours.
4. Respect all team members' opinions.
5. No task reassignment without PM approval.
6. Documentation must be updated regularly.
7. All work must follow agreed quality standards.

PROJECT SCOPE MANAGEMENT

The Inventory Management System (IMS) will include the following features:

- User authentication and role-based access
- Add new inventory items
- Update item quantities
- Delete items
- Automatic low-stock alerts
- Supplier management module
- Purchase order tracking
- Inventory reporting
- Dashboard overview
- Stock history logs
- Export reports (PDF/Excel)
- User management
- Audit trail
- Data backup functionality

REQUIREMENTS

FR= Functional Requirement **NFR**= Non-Functional Requirement

BR=Business Requirement **TR**= Technical Requirement

Req ID	Requirement Description	Type	Source	Priority	Test Case ID	Status
FR1	The system shall allow users to log in securely.	Functional	Project Sponsor	Must	TC01	Planned

FR2	The system shall allow authorized users to add new inventory items.	Functional	Inventory Manager	Must	TC02	Planned
FR3	The system shall allow updating item quantities.	Functional	Inventory Manager	Must	TC03	Planned
FR4	The system shall allow deletion of obsolete items.	Functional	Inventory Manager	Must	TC04	Planned
FR5	The system shall generate low-stock alerts.	Functional	Inventory Manager	Must	TC05	Planned
FR6	The system shall generate inventory reports.	Functional	Finance Officer	Must	TC06	Planned
FR7	The system shall allow supplier registration.	Functional	Procurement Officer	Should	TC07	Planned
FR8	The system shall record purchase transactions.	Functional	Procurement Officer	Should	TC08	Planned
FR9	The system shall maintain stock history logs.	Functional	IT Administrator	Should	TC09	Planned
FR10	The system shall support user role management.	Functional	Project Sponsor	Must	TC10	Planned
FR11	The system shall allow report export to PDF.	Functional	Finance Officer	Could	TC11	Planned
FR12	The system shall provide dashboard analytics.	Functional	Project Sponsor	Should	TC12	Planned
NFR1	The system shall respond within 3 seconds.	Non-Functional	Project Sponsor	Must	TC13	Planned
NFR2	The system shall be available 95% of operational hours.	Non-Functional	Project Sponsor	Must	TC14	Planned
NFR3	The system shall ensure data encryption.	Non-Functional	IT Administrator	Must	TC15	Planned
NFR4	The system shall support up to 50 concurrent users.	Non-Functional	IT Administrator	Should	TC16	Planned
BR1	The system shall reduce stock shortages by at least 30%.	Business	Project Sponsor	Must	TC17	Planned

BR2	The system shall improve reporting efficiency.	Business	Finance Officer	Should	TC18	Planned
TR1	The system shall use a centralized database.	Technical	Lead Developer	Must	TC19	Planned
TR2	The system shall support automated daily backups.	Technical	IT Administrator	Should	TC20	Planned

MoSCoW PRIORITIZATION

MUST HAVE

Login, Add/Edit/Delete Items, Low-Stock Alerts, Reports, User Roles, Database, Testing

SHOULD HAVE

Dashboard analytics, Supplier management, Export reports

COULD HAVE

Advanced analytics, Multi-branch support

WON'T HAVE (This Release)

Mobile application
AI demand forecasting
Integration with external Enterprise Resource Planning(ERP)

ACCEPTANCE CRITERIA

- Low-stock alert must trigger when stock reaches predefined threshold.
- Reports must generate within 3 seconds.
- User login must authenticate valid credentials only.
- System must prevent unauthorized access.

DELIVERABLES

- Approved Project Charter
- RACI Chart

- Team Charter
- Fully functional IMS software
- User Manual
- Test Report
- Deployment Guide
- Final Project Report

OUT OF SCOPE

- Mobile application development
- Integration with third-party ERP systems
- Multi-language support
- AI-based inventory forecasting
- Cloud hosting infrastructure

CONSTRAINTS

- Limited project timeline
- Limited academic budget
- Team availability constraints
- Technology access limitations

ASSUMPTIONS

- Stakeholders will provide timely feedback.
- Required software tools are available.
- Users have basic computer literacy.
- Internet access is stable during development.